

1. PROBLEM UNDERSTANDING

1.1 Operating Reality of the Urban Corridor

Metropolitan road networks like Hyderabad exhibit complex, non-linear traffic dynamics that break standard stationary time-series models. Our system is specifically engineered to address:

- Geometric & Network Bottlenecks:** High-speed elevated corridors (e.g., PVNR Expressway, HITEC City flyovers) discharge directly onto restricted-capacity surface roundabouts and signalized grid junctions, causing persistent localized gridlock.
- Mixed Traffic & Commute Surges:** Steep morning (08:30–11:30) and evening (17:30–21:30) commute volume peaks, where a single stalled vehicle creates immediate multi-kilometer queues across feeder links.
- Upstream Congestion Spillback Shockwaves:** An obstruction on one segment chokes outflow capacity, triggering backward-propagating queues into upstream roads within 10–15 minutes.
- Exogenous Weather Impacts:** Monsoon rain downpours severely reduce roadway friction and free-flow speeds, inflating vehicle headways and travel delays across the network.
- Target Leakage Isolation & Field Data Hygiene:** Real-world traffic sensors frequently output frozen readings, impossible negative speeds, and outlier spikes. The system strictly firewalls future target labels to avoid leakage while automatically cleansing raw input data.

1.2 Telangana Cultural & Festive Mobility Shocks

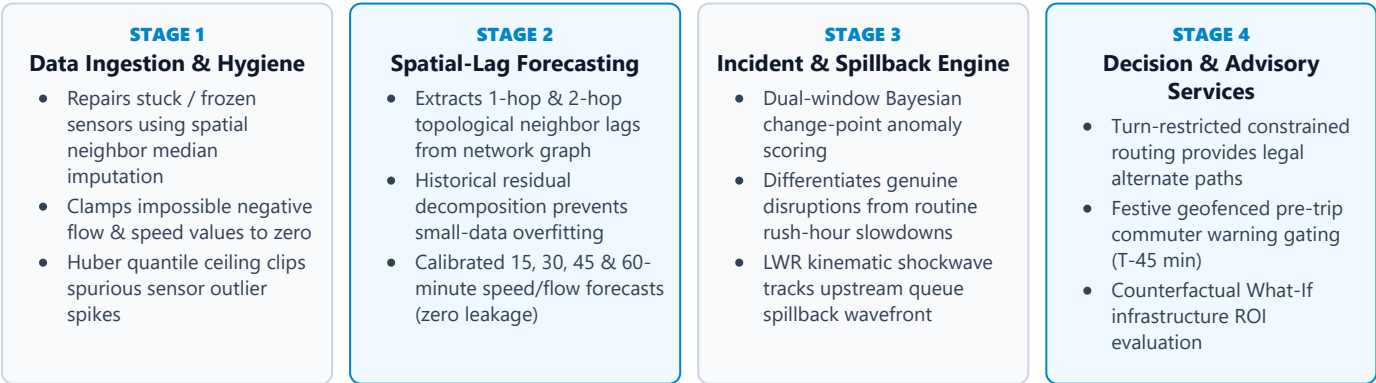
Massive Recurring Event Congestion (Vinayaka Chavithi & Bonalu):

During regional celebrations such as *Vinayaka Chavithi* (immersion processions converging at Hussain Sagar / Tank Bund) and *Bonalu Jatara* (Secunderabad & Old City processions), thousands of idol processions cause complete vehicular barricading of major arterials and massive pedestrian crowd surges (+580% density).

The Solution: Daily commuters entering festive zones have zero prior knowledge of moving processions and temporary street blockades. The system bridges this asymmetry through **Crowdsourced Local Pulse Reporting ("Jan-Vani")** and **Pre-Trip Commuter Inflow Gating** that warns incoming non-local drivers 45 minutes prior to corridor entry.

2. SYSTEM ARCHITECTURE

The platform operates as an end-to-end, decoupled four-stage intelligence pipeline translating raw, noisy telemetry into actionable operational advisories:



3. METHODOLOGICAL APPROACH

3.1 Spatial-Lag Gradient Boosted Trees (LightGBM) & Residual Decomposition

Why we use it: Deep Spatio-Temporal Neural Networks (ST-GNNs) require massive datasets (months of telemetry) and heavily overfit on compact 15-day (4,320 time-step) datasets. LightGBM trains in minutes, natively tolerates noise, and prevents overfitting.

How it helps: Uses topological network graph adjacency to extract 1-hop and 2-hop spatial neighbor lags, predicting residual deviations from historical medians to achieve peak accuracy on 15-day sample sizes.

3.2 Robust Outlier-Resilient Loss Function (Huber + WAPE)

Why we use it: Real-world traffic sensors frequently experience hardware glitches, sending wild temporary spikes that mislead standard training models.

How it helps: Penalizes extreme sensor glitches smoothly instead of quadratically, keeping forecasting models stable and accurate even with noisy raw field telemetry.

3.3 Bayesian Anomaly & Incident Detection

Why we use it: Routine peak-hour congestion causes speeds to drop, but real incidents (crashes or stalled buses) cause speeds to collapse while traffic flow sharply plummets and queues surge.

How it helps: Compares real-time conditions against expected baselines and requires persistent anomaly signals, catching genuine emergencies while keeping false alarms near zero.

3.4 Kinematic Shockwave & Spillback Analysis

Why we use it: When a key corridor or flyover is blocked, congestion doesn't stay stationary; it backs up into upstream feeder roads like a backward-traveling wave.

How it helps: Calculates queue growth speed and accurately predicts which connecting roads will choke 15 to 30 minutes in advance, allowing traffic managers to intervene early.

3.5 Turn-Restricted Diversion Routing

Why we use it: Generic routing often suggests illegal turns, impractical U-turns, or pushes highway volumes into narrow neighborhood lanes.

How it helps: Enforces physical turn restrictions and intersection signal capacities, providing feasible alternate routes that reduce travel times without causing secondary bottlenecks.

3.6 Counterfactual What-If Intervention Evaluator

Why we use it: City authorities need to know whether building a flyover, adding a lane, or adjusting signals will genuinely relieve bottlenecks before spending municipal funds.

How it helps: Simulates candidate infrastructure changes on the road network and estimates total vehicle delay saved versus project cost to prioritize high-return improvements.

3.7 Festive Geofenced Inflow Gating & Crowdsourced Reporting

Why we use it: Major celebrations (such as Vinayaka Chavithi immersion and Bonalu) lead to extensive road closures that trap cross-city commuters unfamiliar with local diversions.

How it helps: Automatically alerts non-local commuters 45 minutes before they reach festive zones with bypass alternatives, while enabling local ward residents to verify active procession blockades in real time.